

Web Development Learning Roadmap (Beginner to Intermediate)

This roadmap is designed for students who are completely new to web development and want to progress systematically from building simple web pages to developing full-stack MERN applications.

1. HTML (Beginner)

HTML (HyperText Markup Language) is the foundation of every website. It is used to structure content such as headings, paragraphs, images, links, forms, and tables.

Resources

Reading

- https://www.w3schools.com/html/html_intro.asp

A beginner-friendly introduction to HTML covering all fundamental concepts with examples and exercises.

Video

- <https://youtu.be/G3e-cpL7ofc?si=fBD4Nb0qV5uw6jlk>

A comprehensive HTML and CSS course that explains concepts visually and helps build a strong foundation.

2. CSS (Beginner)

CSS (Cascading Style Sheets) is used to style and design web pages.

Resources

Reading

- https://www.w3schools.com/css/css_intro.asp

A beginner-friendly guide covering all major CSS concepts.

Video

- <https://youtu.be/G3e-cpL7ofc?si=fBD4Nb0qV5uw6jlk>

Refer to the CSS sections of the same course used for HTML.

Practice Resources

Flexbox

- <https://flexboxfroggy.com/>

An interactive game that makes learning Flexbox layouts fun and intuitive.

CSS Grid

- <https://cssgridgarden.com/>

An interactive game designed to help you understand CSS Grid layouts through hands-on practice.

3. Vanilla JavaScript (Beginner → Intermediate)

JavaScript brings interactivity and dynamic behavior to websites. It is the most important technology to master before moving to React or backend development.

Resources

Beginner Reference

- https://www.w3schools.com/js/js_intro.asp

A simple introduction to JavaScript syntax and fundamentals.

Intermediate Reference

- <https://javascript.info/>

One of the best JavaScript learning resources available online. Covers concepts in depth and provides detailed explanations.

Video Series

- <https://www.youtube.com/playlist?list=PLlasXeu85E9cQ32gLCvAvr9vNaUccPVNP>

Namaste JavaScript by Akshay Saini. Highly recommended for understanding JavaScript concepts and how JavaScript works internally.

DOM Manipulation

- <https://javascript.info/dom-nodes>

A dedicated section covering how JavaScript interacts with HTML elements and web pages.

4. React (Intermediate)

React is a JavaScript library used to build modern user interfaces and single-page applications.

Resources

Beginner Reference

- https://www.w3schools.com/React/react_intro.asp

A beginner-friendly introduction to React fundamentals.

Official Documentation

- <https://react.dev/learn>

The official React learning resource. Recommended for gaining a deeper understanding of React concepts and best practices.

Video Series

📺 [Learn React JS - Full Course for Beginners - Tutorial 2019](#)

A structured React playlist by Codevolution that explains concepts with practical examples.

5. Node.js & Express (Backend Development)

Node.js allows JavaScript to run on the server, while Express simplifies backend development and API creation.

Resources

Video Series

- <https://www.youtube.com/playlist?list=PLCQ35r0Zy6o0tLz9Nq9QzN9iWjK8y9B8x>
- 📺 [Intro to Backend Web Development – Node.js & Express Tutorial for Beginners](#)

A structured introduction to backend development using Node.js.

Reading

- https://www.w3schools.com/nodejs/nodejs_frameworks.asp

An overview of Node.js and backend frameworks.

6. MongoDB (Database)

MongoDB is a NoSQL database commonly used in MERN applications to store and manage application data.

Resources

Reading

- https://www.w3schools.com/mongodb/mongodb_get_started.php

A beginner-friendly guide to MongoDB fundamentals and database operations.

7. Full MERN Stack (Intermediate)

MERN stands for MongoDB, Express, React, and Node.js. This stack enables developers to build complete web applications using JavaScript throughout the entire application.

Resources

Reading

- <https://www.geeksforgeeks.org/mern/understand-mern-stack/>

A good overview of how the MERN stack works and how its different technologies fit together.

Video

- <https://youtu.be/F9gB5b4jgOI?si=SNWWWhja8sd75LKjA>

A complete MERN stack walkthrough demonstrating how frontend, backend, and database components work together in a real application.

8. Git & GitHub Basics

Git is a version control system used to track changes in code, while GitHub is a platform used to host and collaborate on Git repositories.

Resources

Reading

- https://www.w3schools.com/git/git_intro.asp

A beginner-friendly introduction to Git and GitHub workflows.